IEOR 4404 Simulation **Fall 2025**

Practice Final

Instructor: Henry Lam

Problem 1 To estimate θ, we generate 30 independent and identically distributed values each having mean θ. The sample mean is 122.85 and the sample standard deviation is 16.8.

(a) Construct a 95% confidence interval for θ.

(b) Do you think your answer in part (a) is justified? Explain briefly how you justify that this is correct, or alternately what issues it has.

(c) How many additional random values do you need to generate if we want to be 95% confident that our final estimate of θ is correct to within ± 0.5?

Problem 2 Messages arrive at a communication facility according to a time inhomogeneous Poisson process with intensity function $\lambda(t) = 5 + 3t$ for $0 \leq t \leq 5$, and $20 - 2(t - 5)$ for $5 \leq t \leq 10$. The facility consists of one channel. An arriving message will either go to this channel if it is free, in which case the channel becomes busy, or else will be lost if the channel is busy. The amount of time X that a message occupies a channel satisfies

$$P(X > s) = e^{-\sqrt{s}}, s \geq 0,$$

independent of everything else. Suppose that the system is empty at time 0. In this problem, assume your computer only has the capability to generate $Unif(0, 1)$ variables.

(a) Write pseudo-codes to generate the arrivals of messages from time 0 to 10, using both the interarrival time generation approach and the conditional representation approach.

(b) We are interested in estimating $E[L]$, where L is the total number of messages that are lost from time 0 to 10. Explain how you would adapt the pseudo-codes in part (a) (for both approaches therein) to simulate one replication for estimating $E[L]$. In explaining your adaptation, you should also describe how you simulate the random variable X.

(c) Let T_B be the total amount of time in the interval $[0, 10]$ that the channel is busy. Suppose, only for this part, that instead of a time inhomogeneous Poisson process we have a time homogeneous Poisson process with rate λ to generate the message arrivals. What is the conditional expectation $E[L \mid T_B]$?

(d) Back to using the time inhomogeneous Poisson process, let N_B be the number of time intervals during $[0, 10]$ in which the channel is busy, and let those intervals be denoted by $I_1 = [a_1, b_1), I_2 = [a_2, b_2), \ldots, I_{N_B} = [a_{N_B}, b_{N_B})$. Derive an expression for the conditional expectation $E[L \mid N_B, I_1, I_2, \ldots, I_{N_B}]$ in terms of $N_B, a_1, \ldots, a_{N_B}, b_1, \ldots, b_{N_B}$.

(e) By using your answer in part (d), explain how you would modify your answer in part (b) to a conditional Monte Carlo method to estimate $E[L]$.

Problem 3 We run simulation on Y to estimate our target quantity $E[Y]$.

(a) Suppose we use Z_1 as a control variate, and Y and Z_1 are positively correlated. For concreteness, one replication of the control variate estimator is then $W = Y - \beta_1(Z_1 - \mu_1)$, where $\mu_1 = E[Z_1]$ is known. If we want to reduce the variance of naive Monte Carlo, should we choose β_1 to be positive, negative, or dependent on the problem? Explain your answer.

(b) Suppose now we use two control variates Z_1 and Z_2, so that both the pair Y and Z_1, and the pair Y and Z_2, are positively correlated. Concretely, one replication of the control variate estimator is now $W = Y - \beta_1(Z_1 - \mu_1) - \beta_2(Z_2 - \mu_2)$, where $\mu_1 = E[Z_1]$ and $\mu_2 = E[Z_2]$ are known. Provide an example where $\beta_1 > 0$ and $\beta_2 < 0$, but $Var(W) < Var(Y)$.

(c) Now suppose that Z_1 and Z_2 are independent. Can you still find an example as in part (b)? Explain briefly.

Problem 4. Suppose we have the capability to generate $Exp(1)$ variable (e.g., by the inverse transform method).

(a) We are interested in estimating $E[(X+1)^2]$, where X follows a gamma distribution with parameters $\alpha = \frac{3}{2}$ and $\beta = 1$, i.e., $Gamma(\alpha, \beta)$ which has a density

$$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

Write a pseudo-code that only draws $Exp(1)$ to generate one output replication to estimate $E[(X+1)^2]$.

(b) Explain why the estimator you propose in part (a) is valid, including checking any needed conditions.

(c) Based on the estimation variances, do you think the estimator in part (a) is better than the one that directly generates $Gamma(\alpha, \beta)$? Explain with calculations.